

Choral Piece Finder

Author: Axel Riben, Uppsala University **Course:** 5LN712 Information Retrieval **Repository:**

<https://github.com/axelriben/choral-piece-finder> **Demo video:**

<https://youtu.be/GcQV8TLnoyQ>

The project description, system architecture, and installation instructions are in the repository's README .md. This report focuses on what makes the project worth a distinction grade, honest limitations, and reflection on working with AI tools.

What's novel

The project unifies two complementary choral score repositories — the Choral Public Domain Library (CPDL) and Swedish Musical Heritage (SMH) — into a 1,351-work index covering five composers, and exposes it through an LLM-driven agent with eight tools. Four IR contributions distinguish it from a mechanical database wrapper.

Orphan-media discovery beyond catalog navigation. Some PDFs hosted on SMH's server are not linked from any catalog page. A probe enumerates media IDs 1–5000, finds 2,146 files in the dense region (M1–M2500), and matches them against crawled records by composer surname and normalized title. For the project's composers, seven additional editions surface that the source's own catalog does not expose — Stenhammar's *Vårnatt* alone gains four. This illustrates the brief's framing that extending with actions counts as IR: the HEAD-request probing is an action, but it constructs IR-relevant context the source itself does not surface.

Coverage-aware tiered retrieval. The agent reports honestly whether a score is freely downloadable, catalog-only, or unknown. Coverage correlates with copyright status: Stenhammar (PD since 1998) has 56% media coverage, Peterson-Berger (PD since 2013) 31%, Lindberg (PD since 2026) effectively 0%, and Alfvén (copyrighted until 2031) 0%. Lindberg's case is particularly interesting since his works only just entered the public domain this year — coverage is actively evolving.

Cross-source metadata reconciliation (architecturally supported). The unified schema records per-field provenance to flag source discrepancies. For example, CPDL labels Stenhammar's *Vårnatt* as SATB while SMH correctly identifies it as SSAATTBB with piano. In v1 this is unexercised, since the two corpora cover disjoint composers, but the architecture supports activation through IMSLP expansion.

Homonymy disambiguation. When a composer writes multiple pieces with the same title (Peterson-Berger has multiple works titled *Stämning* / *Stemning*), a dedicated tool returns candidates with distinguishing fields (incipit, text author, voicing).

Limitations

Cross-orthography title matching (*Stämning* and *Stemning* as equivalent spellings) is not handled in v1. The agent's summarization layer occasionally introduces factual errors when consolidating multiple disambiguated records, even when the underlying tools return correct structured data. Music21's pitch interpretation for Renaissance editions using original-clef notation produces a systematic octave displacement; v1 includes a heuristic post-processor that

corrects the most common case but not compound errors. Vocal-range analysis additionally depends on MusicXML availability, currently effectively limited to Palestrina. All limitations are documented in the repository.

Reflection on Working with AI Tools

Throughout this project I used Claude to help me plan and Claude Code to help me execute, though I ran into their limitations frequently. The musical nature of the project meant that many important aspects, including pitch and instrumentation, were not reducible to text that an AI can easily understand. This necessitated a lot of manual checking on my part to verify that Claude (and the agent we were building) were not hallucinating musical metadata. The AI agent itself — distinct from Claude, which I used as a development collaborator — used Llama 3.3 70B, which has its own quirks, including making up voicings and numbers, giving up mid-retrieval, and failing to find entries in the database. These issues were softened by adding context to the system prompt, but could not be eliminated entirely.

When the agent did not produce the desired output, Claude developed tunnel vision on extinguishing individual fires but lost the overarching picture, creating new problems while trying to solve others. This required that I pulled the reins and made sure we didn't lose track of the bigger picture. As context grew, Claude increasingly came up with inaccurate diagnoses; I developed a discipline of verifying assumptions with SQL queries before acting on them. The combination of confident AI assertions and large context size meant a single prompt could consume nearly half the tokens provided by Claude Pro in a session.

References

- CPDL. (n.d.). *Choral Public Domain Library*. <https://www.cpdll.org>
- Cuthbert, M. S., & Ariza, C. (2010). music21: A toolkit for computer-aided musicology and symbolic music data. In *Proceedings of the 11th International Society for Music Information Retrieval Conference (ISMIR 2010)* (pp. 637–642).
- Swedish Musical Heritage. (n.d.). *Swedish Musical Heritage* [Levande Musikarv]. <https://www.swedishmusicalheritage.com>